Similarity and normalisation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make each mathematical comparison answer a named question.

Use full raw/unit/cosine tables and explicit zero reports;keep per-vector normalization distinct from fitted preprocessing.

Keep one habit: Assess independent winners and unit distances with checked keys and require evidence beyond a direction match for task claims.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Calculate **vector length** and normalise a nonzero vector

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The squared distance is (3 - 3) squared + (4 - 0) squared = 16. It describes closeness under this representation and rule; it does not prove equivalent meaning.

Opening retrieval

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Look for: Look up a vector using a token ID and preserve sequence shape

Reference example, not the only acceptable wording: Teacher explanation: Vectors are [[1, 1], [0, 1], [0, 2], [0, 0]], mask [1, 1, 1, 0] and shape 4 by 2.

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a transformation preserves the task label

Reference example, not the only acceptable wording: Teacher explanation: Original d = -2 and class 0. Row swapping leaves d = -2 and class 0. Column swapping gives rows [1, 0], [1, 0], d = 2 and class 1.

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a local weighted sum at each valid image position

Reference example, not the only acceptable wording: Teacher explanation: Window rows are [2, 0] and [1, 3]. Products are 2, 0, 0 and 3, giving **output** 5.

Pictured end goal

By the end, I can explain similarity and normalisation using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Embeddings and neighbourhoods

Augmentation and leakage

Convolution as a local operation

For the compact alternative schedule, use two 45-minute sessions. Supply calculators and connect **vector length** to the right-triangle calculation 3 squared plus 4 squared equals 5 squared. Students should justify a rule by the information it retains.

The supplied vectors are constructed examples; no semantic or accuracy claim is established by their geometry.

Outcomes

Calculate **vector length** and normalise a nonzero vector

Compare squared distance with **cosine similarity** and explain a changed ranking

Handle zero vectors explicitly and justify what information a comparison keeps

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach comparison choice,normalization and unavailable direction together

> Keep the original A/B/C pool,pictured P/Qnames and independent P/R/S vectors separate.

Main definitions and complete rankings

Last week, a neighbourhood depended on a chosen distance rule. This week, compare the same fictional vectors in two ways. Squared distance rewards nearby coordinates.

Cosine similarity compares direction and ignores positive changes in length. Neither rule is automatically right for every task.

For Q, length = square root of (3 squared + 4 squared) = square root of 25 = 5. Divide both components by 5 to get its **unit vector** [0.6, 0.8]. Its length is one because 0.6 squared + 0.8 squared = 1.

A becomes the same **unit vector**. C becomes [-0.6, -0.8], pointing in the opposite direction.

These supplied coordinates are for arithmetic practice. They are not learned language embeddings. Normalising each row keeps its direction but removes its original length unless that length is saved separately.

Multiply matching components and add to obtain the dot product. **Cosine similarity** equals dot(Q, candidate) divided by (length(Q) times length(candidate)). Higher cosine means a closer direction.

For A, the dot product is 3 times 6 + 4 times 8 = 50, so cosine = 50 / (5 times 10) = 1. For B it is 9 / 15 = 0.6; for C it is -25 / 25 = -1.

Trace before running: A gives 1, B gives 0.6 and C gives -1. This classroom code assumes finite, moderate numeric **input**s. It rejects empty, unequal-length and zero vectors.

Keep full precision for decisions; rounding here is only for display. Cosine lies between -1 and 1 in exact arithmetic and is not a probability.

Main query Q[3, 4], length 5 · Candidate length · Raw squared distance · Cosine:larger wins · Unit-vector squared distance:smaller wins

> **Main query Q[3, 4], length 5:** A[6, 8]
>
> **Candidate length:** 10

> **Raw squared distance:** 25
>
> **Cosine:larger wins:** 1
>
> **Unit-vector squared distance:smaller wins:** 0

> **Main query Q[3, 4], length 5:** B[3, 0]
>
> **Candidate length:** 3
>
> **Raw squared distance:** 16
>
> **Cosine:larger wins:** 0.6
>
> **Unit-vector squared distance:smaller wins:** 0.8

> **Main query Q[3, 4], length 5:** C[-3, -4]
>
> **Candidate length:** 5
>
> **Raw squared distance:** 100
>
> **Cosine:larger wins:** -1
>
> **Unit-vector squared distance:smaller wins:** 4

> **Main query Q[3, 4], length 5:** Winner among original A/B/C
>
> **Candidate length:** Different criterion
>
> **Raw squared distance:** B
>
> **Cosine:larger wins:** A
>
> **Unit-vector squared distance:smaller wins:** A

Cosine divides a dot product by both nonzero lengths. Compare all original A/B/C candidates under the same rule before ranking. Unit-vector squared distance is2−2 cosine, so it ranks the same way as largest cosine when both sides have unit length. Keep full precision for decisions and round only display values. The code assumes finite moderate numeric inputs and rejects empty, mismatched or zero vectors.

A cosine in[−1, 1]is a mathematical comparison, not a calibrated probability or evidence that records mean the same thing.

Unit-distance identity and guided candidate D

After normalising both Q and every candidate, squared distances to A, B and C are 0, 0.8 and 4. For **unit vector**s, squared distance = 2 - 2 times cosine.

Thus the smallest normalised distance and largest cosine produce the same ranking, including ties, in exact arithmetic.

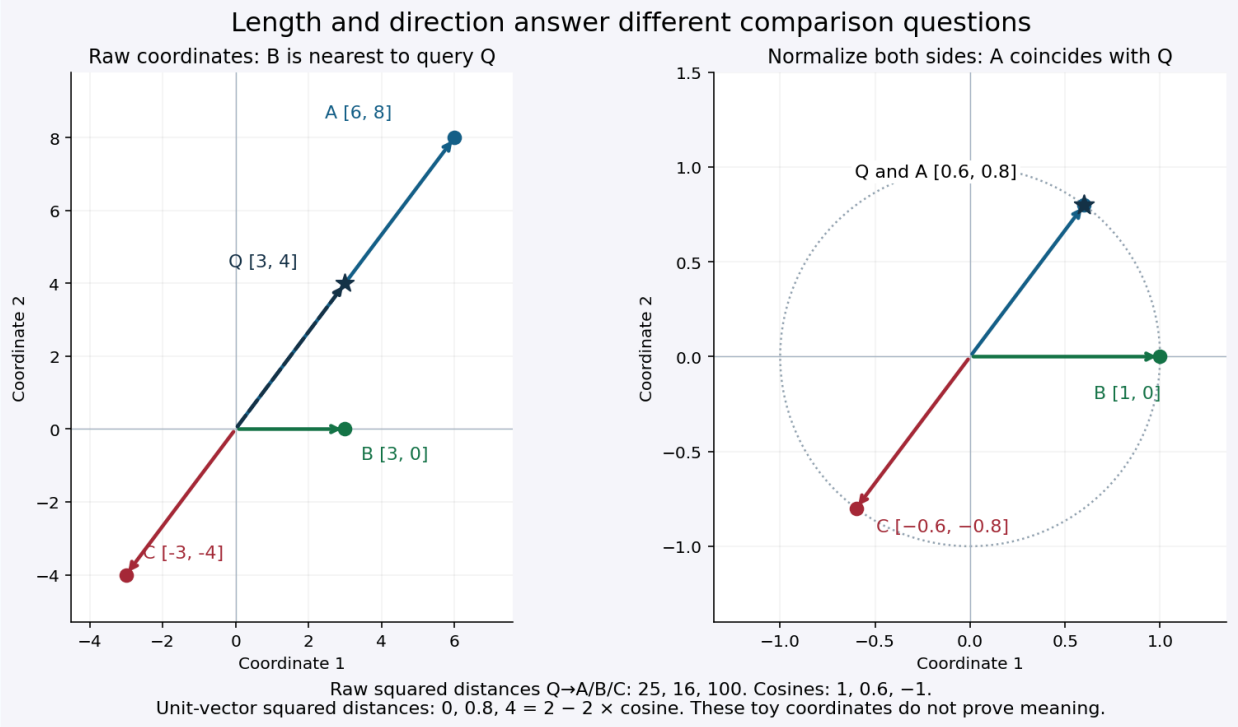
> **Add candidate D [0, 5]. Calculate its raw squared distance and cosine from Q [3, 4]. Compare D with A under each rule.**

Guided: D has squared distance 9 + 1 = 10 and cosine 20 / 25 = 0.8. Raw distance prefers D over A (10 versus 25); cosine prefers A (1 versus 0.8).

Exit: Q and A share direction but have different lengths and may represent different totals. Cosine 1 proves aligned nonzero vectors, not equivalent meaning.

Check the task, retained information, zero and tie policies, and validation performance against a baseline; reserve final evaluation.

Side-by-side geometry and complete zero/magnitude policy



Complete information/policy case	Meaning and declared response
Counts[3, 4]versus[6, 8]	Totals7 and14 but same unit vector[.6, .8]; length normalization loses total magnitude
Keep unit vector plus its original norm	Can reconstruct the exact original vector in exact arithmetic; this does not prove a better model
Zero candidate Z[0, 0]with query[3, 4]	Raw squared distance 25is defined; cosine is undefined. Exclude/report Z under this lesson's cosine policy.
Only zero candidates remain	Return no comparison, not an invented similarity or winner
Zero query	Reject cosine comparison; use a separately declared fallback or review policy
Perpendicular nonzero vector[-4, 3]	Cosine0is valid; do not confuse it with undefined zero-vector cosine
Positive versus negative scaling	Cos(Q, 2Q)=1; cos(Q, −Q)=−1

Complete information/policy case	Meaning and declared response
Per-vector L2 normalization versus fitted feature scaling	This norm uses one vector only; dataset-derived means/scales are different and must use training statistics

Separate pictured query and candidate scope

Compare angular alignment with raw distance on the same vectors.

Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses query[2, 0], candidate P[8, 0]and a candidate named Q[2, 1]. Here Q is a candidate name, not the main query Q[3, 4]. Raw squared distances 36 and1 prefer candidate Q. Cosines1 and2/$\sqrt{5}$≈.8944 prefer P. Normalize both sides:the query and P become[1, 0]. Replacing P with[0, 0]makes its norm 0 and cosine undefined; it does not create a valid perpendicular cosine 0.

These toy rankings do not select a universally best metric.

Trace the undefined zero-vector replacement

AI ACADEMY / AI-5/27 · 02 · TRACE THE CASE

Follow the deciding steps

01 Raw distances

P distance 36; Q distance 1, so Q is closer in raw coordinates.

02 Angular similarity

Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely.

03 Normalise

Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Raw distances	P distance 36; Q distance 1, so Q is closer in raw coordinates.
2. Angular similarity	Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely.
3. Normalise	Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates.

Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

Change: Replace P with [0, 0].

Prediction: Cosine and unit normalisation for P are undefined.

Reason: The zero vector has norm 0, so dividing by its norm is invalid.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Vector length	The square root of the sum of squared components, also called the L2 norm
L2 normalisation	Dividing every component of a nonzero vector by its length
Cosine similarity	The dot product of two nonzero vectors divided by the product of their lengths
Unit vector	A vector with length one

Compare the same candidates using two rules

CANDIDATE	RAW SQUARED DISTANCE	COSINE SIMILARITY
A [6, 8]	25	1
B [3, 0]	16	0.6
C [-3, -4]	100	-1

For Q [3, 4], raw squared distance selects B by minimising the score. Cosine selects A by maximising the score. The candidates stayed the same; the comparison rule changed.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the same candidates using two rules

CANDIDATE	RAW SQUARED DISTANCE	COSINE SIMILARITY
A [6, 8]	25	1
B [3, 0]	16	0.6
C [-3, -4]	100	-1

For Q [3, 4], raw squared distance selects B by minimising the score. Cosine selects A by maximising the score. The candidates stayed the same; the comparison rule changed.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

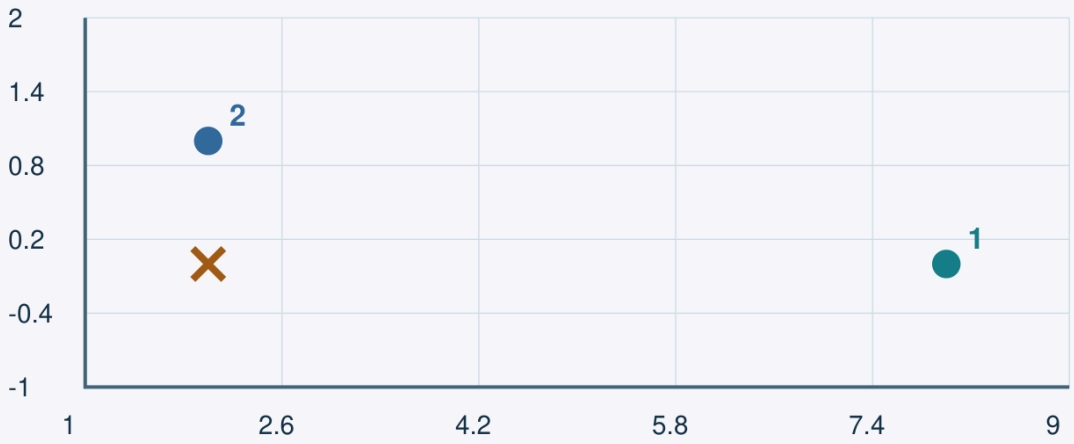
Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Similarity and normalisation

The facts we will use

AI ACADEMY / AI-5/27

Raw proximity differs from angular alignment



Horizontal: first vector coordinate; vertical: second vector coordinate

1: P at (8, 0)

2: Q at (2, 1)

Cross: query at (2, 0)

Distance rule: Compare squared Euclidean distance and cosine

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/27

Similarity and normalisation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Raw distances

P distance 36; Q distance 1, so Q is closer in raw coordinates.



2. Angular similarity

Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely.



3. Normalise

Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

CHANGE ONE THING

Replace P with [0, 0].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule.

Calculate squared distances after normalising both sides and check the relationship to cosine.

Teacher check: Teacher explanation: Raw squared distances are 25, 16, 100, so B wins. Cosines are 1, 0.6, -1, so A wins. Unit-vector squared distances are 0, 0.8, 4, matching 2 - 2 times each cosine and selecting A.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Similarity and normalisation. Proposed explanation: “Cosine is a probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Its possible negative values and geometric definition show that it is not a calibrated probability.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Raw distance includes length differences; cosine compares direction after accounting for norms. P shares direction but has a much larger magnitude. Changed-case check: Cosine and unit normalisation for P are undefined.

The zero vector has norm 0, so dividing by its norm is invalid.

Changed-case reasoning: Cosine and unit normalisation for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a ledger for squared components, length, unit components and score. Allow a calculator and separate lower-is-better distance from higher-is-better similarity.

Extension: expand the squared distance between two **unit vector**s to derive 2 - 2 times their dot product.

For floating-point retrieval, declare a tolerance and retain all scores within it of the best; show that rounding first can create false ties.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve squared distance and explain why a geometric score alone does not establish meaning.
7 to 18 minutes	Compute lengths and unit vectors. Complete W1 with both positive and opposite directions.
18 to 32 minutes	Trace the cosine formula and optional code, checking which score is maximised.
32 to 45 minutes	Use the diagram and W2 to contrast raw and normalised distance rankings.
Session 2 start to 8 minutes	Complete the guided D comparison, retaining the same query and candidate A.
8 to 20 minutes	Discuss event counts and zero vectors. Complete W3 and require an explicit no-comparison outcome.
20 to 35 minutes	Collect W4 independently, including raw scores, cosines, normalised distances and an explanation.
35 to 45 minutes	Complete W5 and the exit response. Give feedback before administering the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main raw winner Bversus cosine/unit-distance winner A; the identity requires both normalized vectors. W3Zexcluded, B cosine.6; zero query rejects; only Z gives no comparison, not cosine 0. W5 normalization loses total counts 7 versus14, and selecting with final outcomes makes development evidence.

Independentquery[5, 0], P[10, 0], R[4, 3], S[0, 5]:raw squared distances 25/10/50 select R; cosines 1/.8/0 select P; unit distances 0/.4/2 select P. The same positive direction can have greater length. Preserve metric, mask/candidate roles, zero policy and tie tolerance.

Diagnose the current misconceptions

Cosine is a probability: Its possible negative values and geometric definition show that it is not a calibrated probability.

Normalisation keeps all information: Positive multiples collapse to the same **unit vector**; preserve length separately if the task needs it.

Any scaling leaves cosine unchanged: Positive scaling preserves direction. Multiplying only one vector by a negative number reverses the cosine sign; zero scaling makes it undefined.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

Explain the changed ranking.

Teacher explanation: Raw squared distances to P, R and S are 25, 10, 50, selecting R. Cosines are 1, 0.8, 0, selecting P. Normalised squared distances are 0, 0.4, 2.

P shares the query direction despite its greater length; raw distance also reflects that length difference.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace the unit vectors

For Q [3, 4], A [6, 8] and C [-3, -4], give each length and **unit vector**. Explain one distinction that normalisation removes and one that it preserves.

Teacher explanation: Lengths are 5, 10 and 5. **Unit vector**s are [0.6, 0.8], [0.6, 0.8] and [-0.6, -0.8]. The length difference between Q and A disappears; the opposite direction of C is preserved.

W2 Compare the rankings

For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

Teacher explanation: Raw squared distances are 25, 16, 100, so B wins. Cosines are 1, 0.6, -1, so A wins. Unit-vector squared distances are 0, 0.8, 4, matching 2 - 2 times each cosine and selecting A.

W3 Handle a vector without direction

A candidate set contains Z [0, 0] and B [3, 0]. With nonzero query Q, what happens under this lesson's zero-vector policy? What if Q is zero, or Z is the only candidate?

Explain why replacing an undefined cosine with zero is misleading.

Teacher explanation: Exclude Z and report it; compare B normally. Reject a zero query. If only Z exists, report no comparison. Zero cosine represents perpendicular nonzero directions, whereas the zero vector has no direction.

W5 Audit a proposed improvement

For event-count vectors [3, 4] and [6, 8], what useful information can normalisation remove? A team chooses cosine because it improves results on the final evaluation set, then calls that same set independent **evidence**.

Explain how to repair the evaluation plan.

Teacher explanation: Normalisation removes magnitude: totals 7 and 14 become indistinguishable in the **unit vector**s. Choose the rule using task needs and **training**/validation **evidence**, preserving a baseline.

A final set used to choose the rule is no longer independent final **evidence**; reserve new untouched evaluation data or explicitly report the limitation.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: D has squared distance 9 + 1 = 10 and cosine 20 / 25 = 0.8. Raw distance prefers D over A (10 versus 25); cosine prefers A (1 versus 0.8).

Exit: Q and A share direction but have different lengths and may represent different totals. Cosine 1 proves aligned nonzero vectors, not equivalent meaning.

Check the task, retained information, zero and tie policies, and validation performance against a baseline; reserve final evaluation.

W1 Trace the unit vectors

W2 Compare the rankings

W3 Handle a vector without direction

W4 Compare a new case independently

Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

Explain the changed ranking.

Teacher explanation: Raw squared distances to P, R and S are 25, 10, 50, selecting R. Cosines are 1, 0.8, 0, selecting P. Normalised squared distances are 0, 0.4, 2.

P shares the query direction despite its greater length; raw distance also reflects that length difference.

W5 Audit a proposed improvement

Independent assessment and fresh retry

W4 requires raw scores 25, 10, 50; cosines 1, 0.8, 0; normalised distances 0, 0.4, 2; and winners R then P. Fresh retry: query [0, 5], candidates D [0, 10], E [3, 4], F [5, 0].

Raw squared distances are 25, 10, 50; cosines 1, 0.8, 0; unit distances 0, 0.4, 2. Raw distance selects E; cosine selects D. Ask students to explain why a zero query is rejected.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled similarity and normalisation **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **vector length** and normalise a nonzero vector.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a ledger for squared components, length, unit components and score. Allow a calculator and separate lower-is-better distance from higher-is-better similarity.

Extension: expand the squared distance between two **unit vector**s to derive 2 - 2 times their dot product.

For floating-point retrieval, declare a tolerance and retain all scores within it of the best; show that rounding first can create false ties.

Week 28 examines transfer learning, comparing a reused representation with a baseline and testing what can transfer to a new task.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/28, ai-5/30, ai-5/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Calculate **vector length** and normalise a nonzero vector

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule.

Calculate squared distances after normalising both sides and check the relationship to cosine.

Reasoned practice answer: Teacher explanation: Raw squared distances are 25, 16, 100, so B wins. Cosines are 1, 0.6, -1, so A wins. Unit-vector squared distances are 0, 0.8, 4, matching 2 - 2 times each cosine and selecting A.

Check every learner before discussion: Use the worked case for Similarity and normalisation. Proposed explanation: “Cosine is a probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Its possible negative values and geometric definition show that it is not a calibrated probability.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace P with [0, 0]. Require a **prediction** and a reason before comparing. Expected result: Cosine and unit normalisation for P are undefined. Reason: The zero vector has norm 0, so dividing by its norm is invalid.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

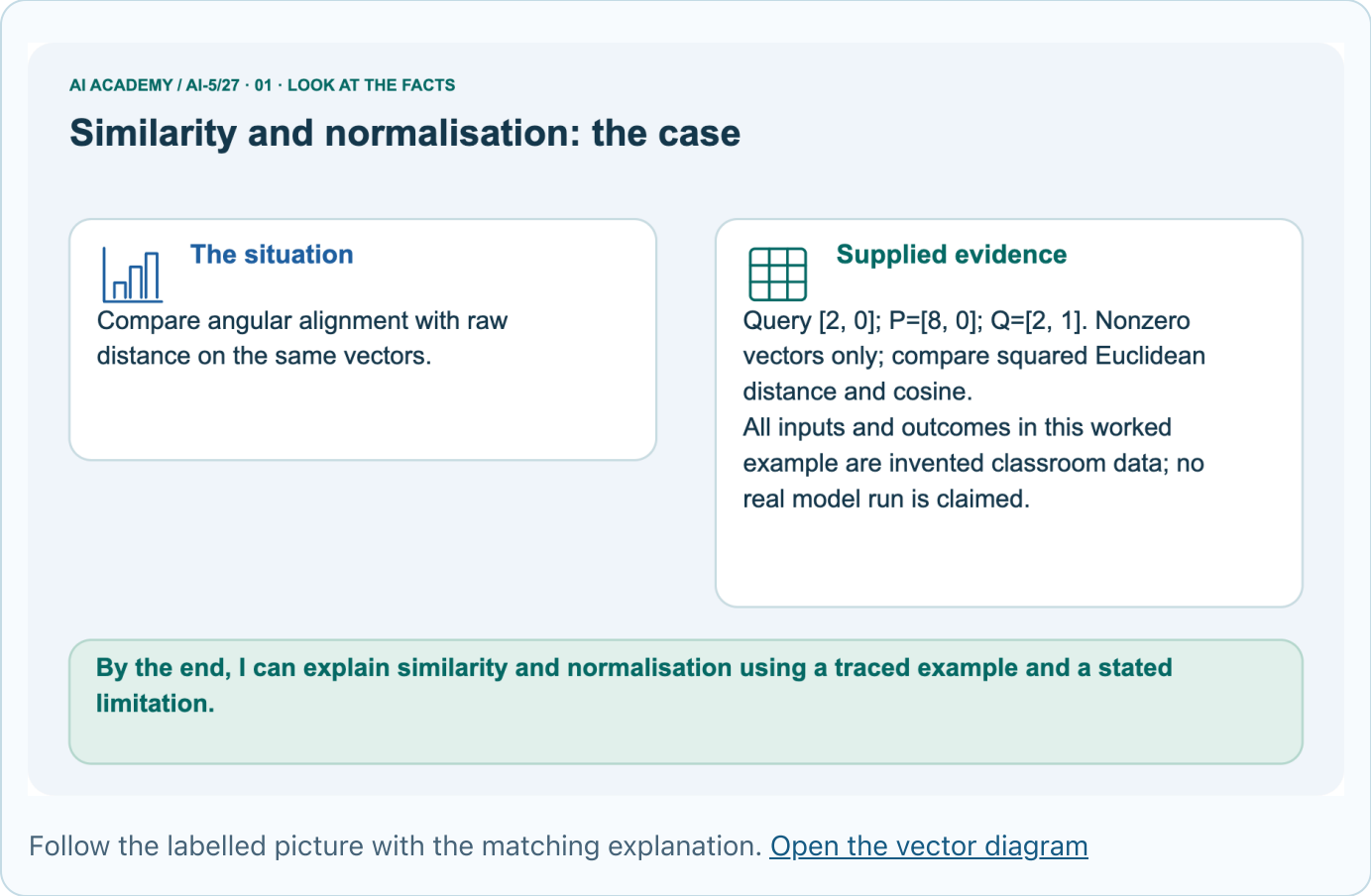
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Similarity and normalisation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

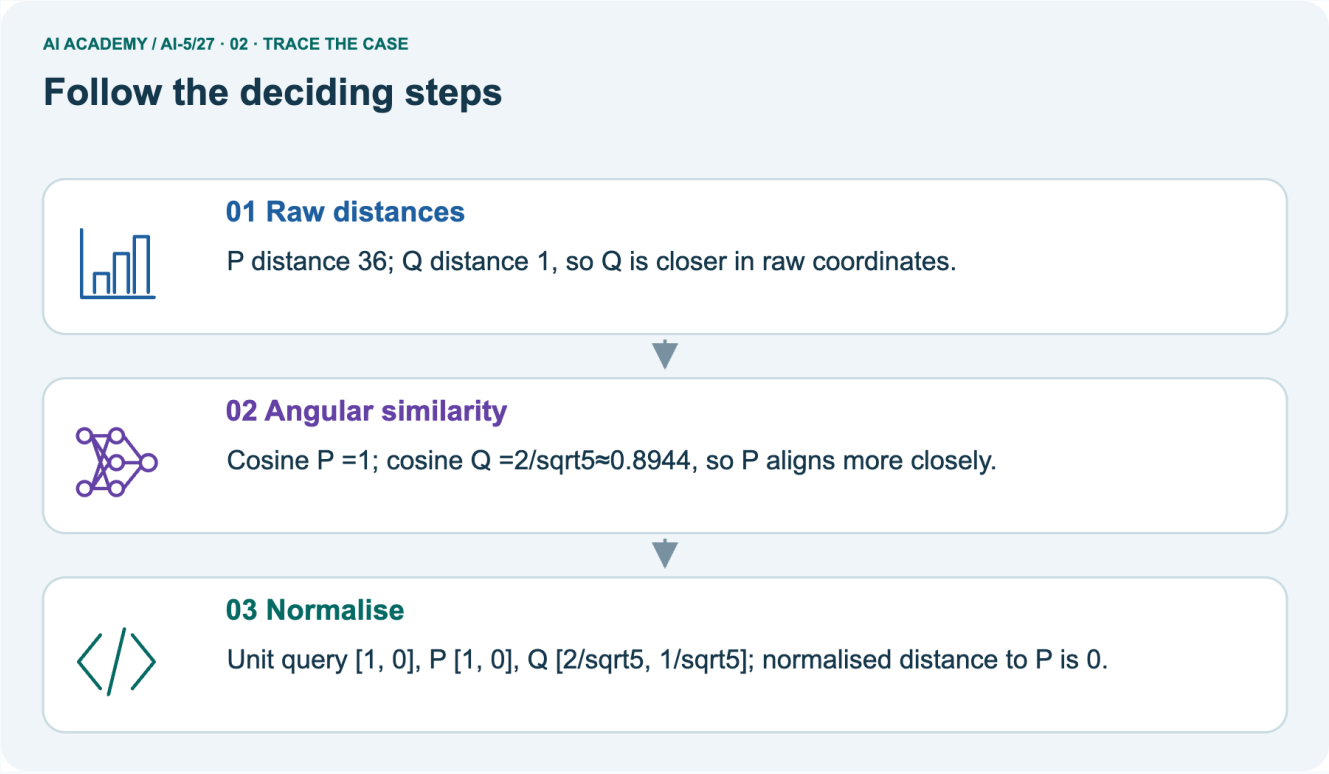
Compare angular alignment with raw distance on the same vectors. Supplied facts: Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



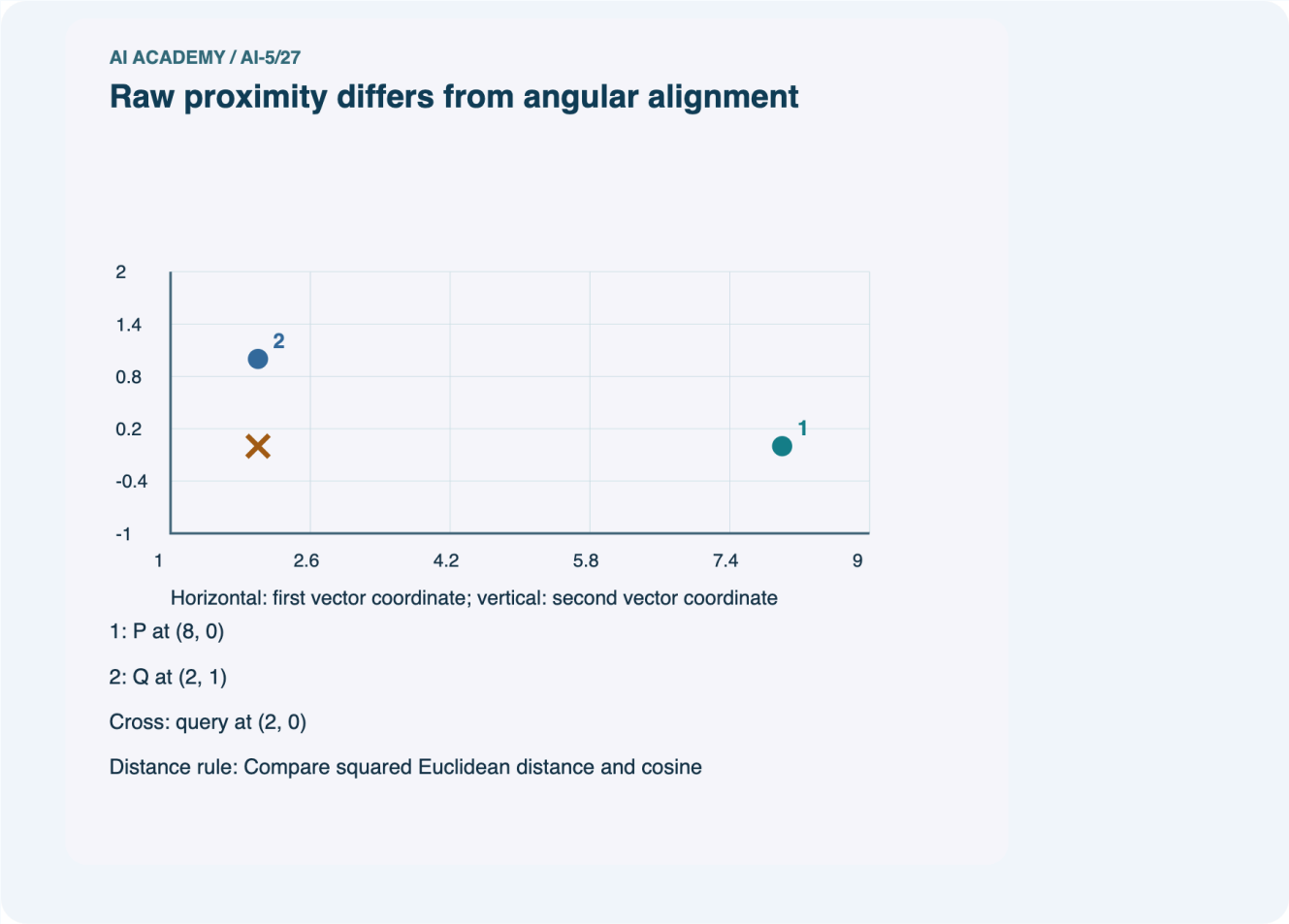
Follow the labelled picture with the matching explanation. Open the vector diagram

Raw distances: P distance 36; Q distance 1, so Q is closer in raw coordinates. → Angular similarity: Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely. → Normalise: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Raw proximity differs from angular alignment



Follow the labelled picture with the matching explanation. Open the vector diagram

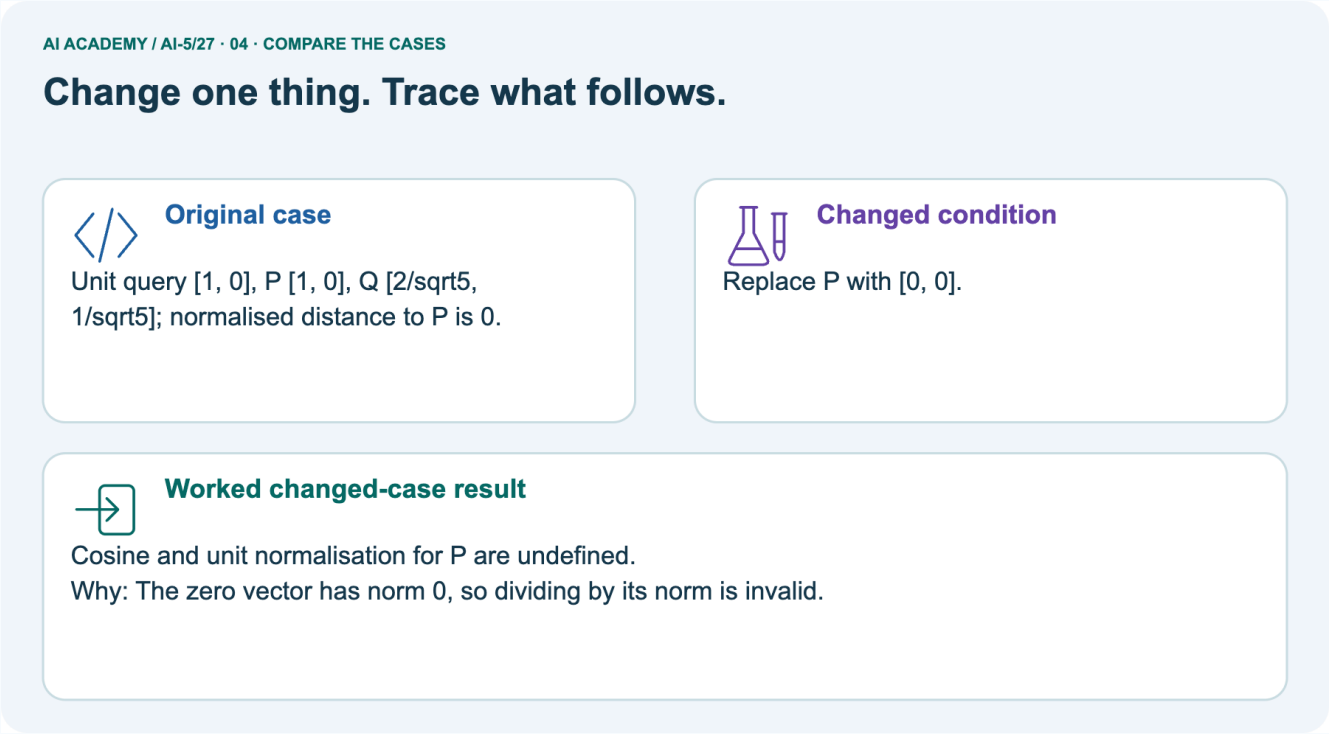
Raw proximity differs from angular alignment. P: (8, 0). Q: (2, 1). Query query: (2, 0). Axes: first vector coordinate; second vector coordinate. Distance: Compare squared Euclidean distance and cosine.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0. Changed condition: Replace P with [0, 0]. Expected result: Cosine and unit normalisation for P are undefined.

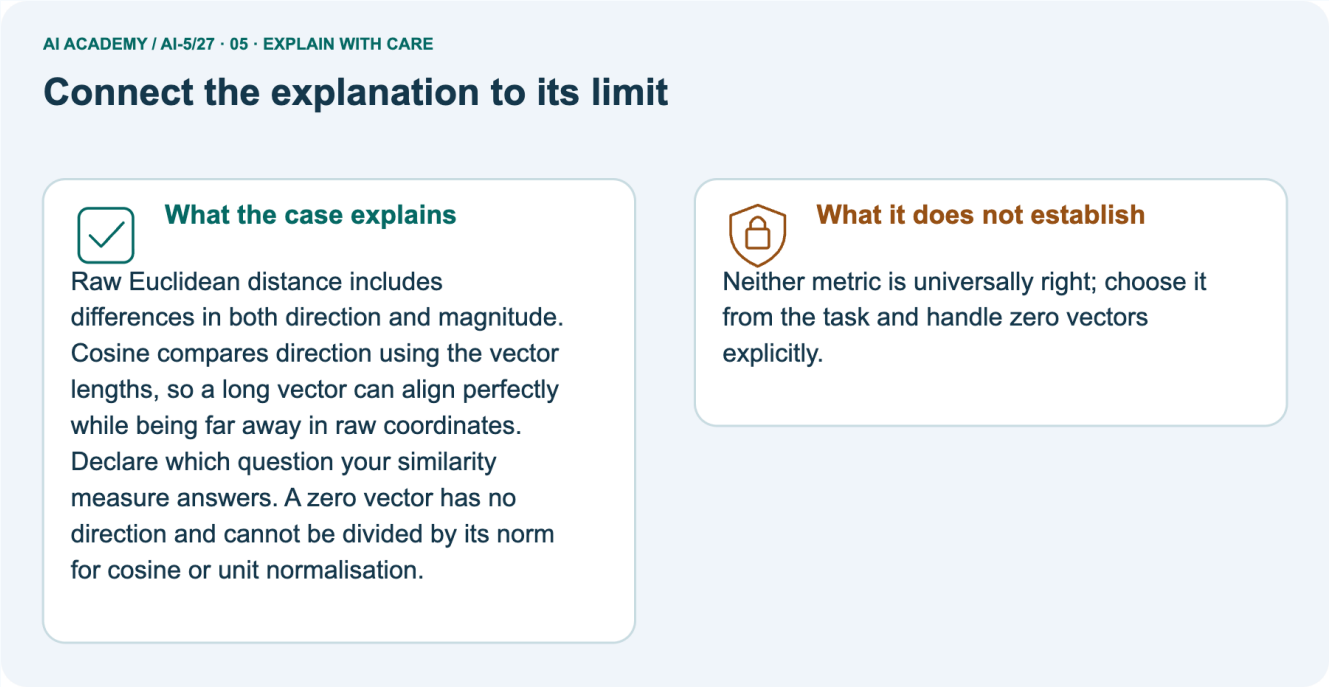
Reason: The zero vector has norm 0, so dividing by its norm is invalid.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The zero vector has norm 0, so dividing by its norm is invalid.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates.

Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

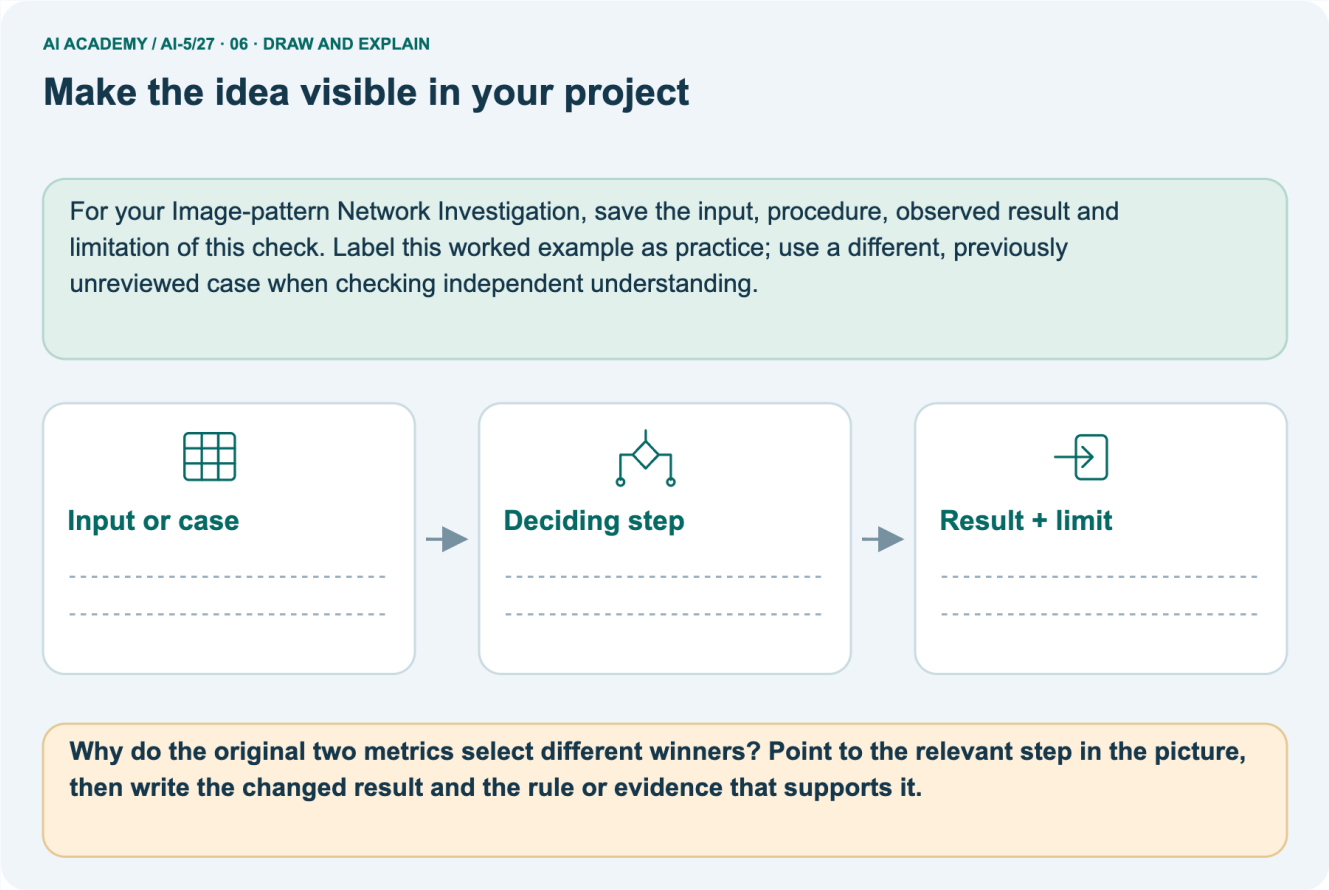
Limit: Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Raw distance includes length differences; cosine compares direction after accounting for norms. P shares direction but has a much larger magnitude.

Changed-case check: Cosine and unit normalisation for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Similarity and normalisation: the case

02. Follow the deciding steps

03. Raw proximity differs from angular alignment

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.